Jacob A. Johnson

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EDUCATION

Ph.D. in Statistics — Expected May 2029

University of Wisconsin-Madison, *Madison, WI*

B.S. in Statistical Science, Minor in Mathematics — June 2024

Brigham Young University, *Provo, UT*

- *summa cum laude* (4.00/4.00 GPA)
- National Merit Scholar

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Statistics Intern — May 2025 – Present

Sandia National Laboratories, *Albuquerque, NM*

- Developed statistical machine learning models to forecast extreme temperatures at subseasonal (2–8 weeks out) timescales across the contiguous United States.
- Evaluated and developed various dimension reduction techniques to extract predictive features from 40+ years of atmospheric data to be used as inputs for temperatures forecasting.

Financial Data Analytics Intern — May 2024 – Aug 2024

Williams International, *Pontiac, MI*

- Created statistical models to forecast inventory using historical production and usage data.
- Developed an anomaly detection program to monitor and identify deviations in expenses, detecting unusual spending patterns and potential fraud across business units.
- Automated forecasting and anomaly detection processes, reducing manual oversight and providing executives with actionable insights.

Research Assistant — May 2022 – April 2024

Brigham Young University, *Provo, UT*

- Utilized Bayesian machine learning techniques to combine climate data products for predicting average monthly precipitation in the Himalayan Mountains.
- Developed new subsampling methods in R to speed up computation of Gaussian Process calculations for large spatial data.

PUBLICATIONS

[1] Matthew J. Heaton and **Johnson, Jacob A.** "Minibatch Markov Chain Monte Carlo Algorithms for Fitting Gaussian Processes". In: *Bayesian Analysis* (2024), pp. 1–23. DOI: 10.1214/24-BA1459.

[2] **Johnson, Jacob A.**, Matthew J. Heaton, William F. Christensen, Lynsie R. Warr, and Summer B. Rupper. "Fusing Climate Data Products Using a Spatially Varying Autoencoder". In: *Journal of Agricultural, Biological and Environmental Statistics* (Oct. 2024). DOI: 10.1007/s13253-024-00657-3.

AWARDS & HONORS

- UW-Madison Statistics Department Outstanding Performance as a TA award 2024 – 2025
- ASA Section on Statistics and the Environment Student Paper Competition Winner 2024

PRESENTATIONS

Oral Presentations

- "Modeling Bayesian Transport Map Uncertainty for Non-Gaussian Spatial Data via Laplace Approximation." Joint Statistical Meetings (JSM), Nashville, TN. August 2025.
- "Climate Data Melding via Spatially varying Bayesian Autoencoder." Joint Statistical Meetings (JSM), Portland, OR. August 2024.
- "Minibatch Markov chain Monte Carlo Algorithms for Fitting Gaussian Processes." CPMS Student Research Conference, Provo, UT. February 2024.
- "Climate Data Melding via Spatially varying Bayesian Autoencoder." Joint Statistical Meetings (JSM), Toronto, ON. August 2023.
- "Climate Data Melding via Spatially varying Bayesian Autoencoder." CPMS Student Research Conference, Provo, UT. February 2023.

Poster Presentations

- "Minibatch Markov chain Monte Carlo Algorithms for Fitting Gaussian Processes." ENVR Workshop, Boulder, CO. October 2024.
- "Climate Data Melding via Spatially varying Bayesian Autoencoder." ENVR Workshop, Provo, UT. October 2022.

TEACHING EXPERIENCE

Teaching Assistant, *University of Wisconsin-Madison* Fall 2024 – Present

- STAT 324 - Introductory Statistics for Engineers (Fall 2024, Spring 2025)
- STAT 240 - Data Science Modeling I (Fall 2025, Spring 2026)

Teaching Assistant, *Brigham Young University* Sept 2021 – April 2022

- STAT 240 - Probability and Inference I (Fall 2021, Spring 2022)

SKILLS

- **Technical:** R, Python, SQL, Linux shell processes.
- **Technical: Languages:** English (Native), Spanish (Fluent).